GIS-Based Flood Risk Assessment of Fire Island, NY: A LiDAR-Derived DEM Analysis from Robert Moses to Watch Hill

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Abstract

As climate change accelerates, rising sea levels and increasingly intense coastal storms have contributed to more frequent and severe flooding in low-lying coastal regions. Fire Island, a barrier island off the southern coast of Long Island, New York, is especially susceptible to flooding due to its narrow width, minimal elevation, and exposure to both the Atlantic Ocean and Great South Bay. The study area, extending from Robert Moses Beach to Watch Hill, is characterized by low relief topography and dynamic coastal processes that heighten its vulnerability to storm surge and tidal inundation.

To evaluate potential flooding scenarios within this region, Geographic Information Systems (GIS) techniques were employed to model and analyze projected inundation patterns. High-resolution Light Detection and Ranging (LiDAR) data were processed using Global Mapper to generate a bare-earth Digital Elevation Model (DEM), removing vegetation, structures, and other surface features to accurately represent ground elevation. This elevation model enabled detailed flood simulations under varying sea-level rise scenarios and storm surge conditions.

Results from the simulations indicate that substantial portions of the study area, particularly low-elevation bayside zones and interior regions closer to the Atlantic shoreline and low-lying interior zones, are at significant risk of inundation. Areas with slightly higher dune elevations provide limited natural protection, though these features are vulnerable to erosion and overwash during major storm events. The findings highlight priority zones for emergency planning and resource allocation, particularly the west end access point of the Robert Moses Causeway.

The GIS-based modeling framework utilized in this study can be adapted for other barrier island systems and coastal environments worldwide. By identifying areas of highest flood susceptibility, this approach supports coastal management strategies, infrastructure planning, and emergency response preparedness in the face of continued sea-level rise.

Introduction

Flooding is widely recognized as one of the most impactful and frequent natural hazards worldwide, producing substantial socio-economic losses and increasing in both frequency and intensity under changing climatic conditions (Tsakiri et al., 2018). Coastal and riverine environments alike are experiencing heightened vulnerability due to the combined effects of sea-level rise, extreme precipitation events, and anthropogenic modifications of natural systems. In particular, low-lying coastal barrier systems, such as Fire Island, New York, represent highly dynamic environments where even modest increases in water levels can lead to widespread inundation and disruption of infrastructure.

Previous research has demonstrated that flood processes are strongly controlled by topography, hydrologic connectivity, and geomorphological constraints. In riverine systems such as the Mohawk River in New York, flooding has been linked to both climatic drivers (e.g., precipitation and storms) and physical obstructions such as bridges, dams, and ice jams, which can restrict flow and elevate upstream water levels (Lakeram et al., 2017; Foster et al., 2011). These findings highlight the importance of understanding not only water input conditions but also landscape controls that influence flood propagation and accumulation. Furthermore, historical analyses show that flooding events can produce severe damage to infrastructure, businesses, and residential areas, emphasizing the need for improved prediction and risk assessment tools (Swan et al., 2016; Sisti et al., 2016).

Advances in geospatial technologies, particularly Geographic Information Systems (GIS) and Light Detection and Ranging (LiDAR), have significantly improved the ability to model and predict flood hazards. High-resolution Digital Elevation Models (DEMs) derived from LiDAR data allow for detailed representation of terrain, enabling accurate simulation of water flow and inundation patterns. LiDAR-based bare-earth models, which remove vegetation and built structures, provide a realistic depiction of ground elevation and are critical for flood simulations, especially in low-relief environments (Foster & Marsellos, 2012). These datasets enable volumetric calculations and incremental water-level simulations, allowing researchers to identify critical flood thresholds and trigger levels with high precision (Foster & Marsellos, 2012; Marsellos et al., 2010).

Numerous studies have applied GIS and LiDAR techniques to reconstruct past flood events and simulate future scenarios. For example, LiDAR-based flood mapping and volumetric analysis have been successfully used to reconstruct ice-jam flooding events and evaluate flood extents under varying conditions in the Mohawk River system (Marsellos et al., 2010; Marsellos et al., 2010b). Similarly, geospatial analyses have demonstrated that high-resolution elevation data are essential for accurately modeling flood dynamics and identifying vulnerable zones, particularly in areas with subtle elevation differences (Marsellos, 2012; Rienzo et al., 2018). These approaches have also been extended to forecasting applications, where time series decomposition, statistical models, and machine learning techniques such as artificial neural networks (ANN) and ARIMA have been used to predict water discharge and flood events with high accuracy (Tsakiri et al., 2014; Tsakiri et al., 2018; Tran et al., 2019; Plitnick et al., 2018).

Beyond hazard mapping, recent studies emphasize the importance of integrating flood modeling with risk management and decision-making frameworks. GIS-based analyses have been used not only to identify flood-prone areas but also to evaluate community preparedness, infrastructure vulnerability, and the need for real-time warning systems (Lakeram et al., 2017; Marsellos et al., 2016). These integrated approaches are particularly important in densely populated or infrastructure-dependent regions, where even partial inundation can lead to significant disruptions in transportation networks and emergency response capabilities.

While much of the existing literature has focused on riverine flooding, similar principles apply to coastal barrier systems, where low elevation, limited topographic relief, and direct connectivity to surrounding water bodies increase susceptibility to inundation. Barrier islands are especially vulnerable because they are subject to both ocean-driven processes (e.g., storm surge, wave overwash) and bay-side flooding, resulting in complex bidirectional flooding patterns. In such environments, small elevation differences can determine whether areas remain protected or become inundated, making high-resolution elevation data essential for accurate risk assessment (Weinstein & Marsellos, 2018).

This study builds upon previous GIS and LiDAR-based flood modeling approaches to evaluate flood risk on Fire Island, New York, with particular emphasis on infrastructure vulnerability. Specifically, it focuses on the region between Robert Moses and Watch Hill, where low-lying terrain and critical access routes create heightened risk during storm events and sea-level rise scenarios.(See Figure 1) By integrating LiDAR-derived DEMs with GIS-based inundation modeling, this research aims to identify spatial patterns of flood susceptibility and determine thresholds at which critical infrastructure—particularly the Robert Moses Causeway—becomes compromised (See Figure 2).

Figure 1: The study area of Fire Island, NY.

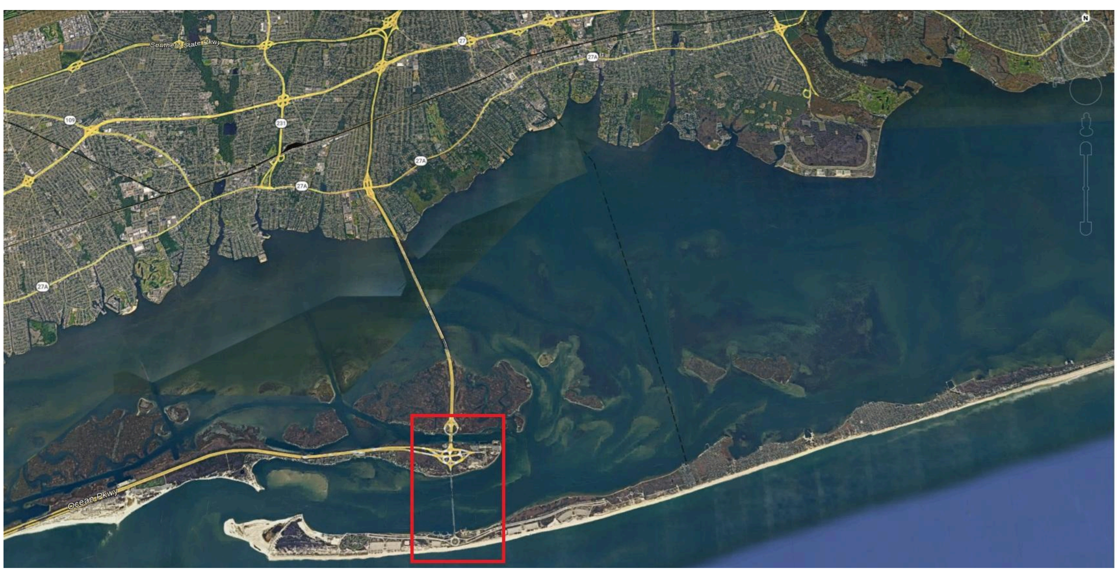

Figure 2: Robert Moses Bridge Infrastructure

In addition to mapping inundation extents, this study advances prior work by explicitly linking flood progression to transportation accessibility and emergency response considerations. Understanding the sequence of infrastructure failure and the conditions under which access routes become impassable is essential for effective evacuation planning and coastal resilience strategies. Ultimately, this research contributes to a growing body of work that leverages geospatial technologies and data-driven modeling to support hazard mitigation and sustainable coastal management.

In addition to identifying areas of inundation, this study places particular emphasis on how flooding impacts transportation accessibility and critical infrastructure, especially the Robert Moses Causeway. The Robert Moses Causeway is the primary vehicular access to Fire Island , which provides land-based access between Fire Island and Long Island. While the east end of Fire Island is accessible via the Smith's Point Bridge, westward travel has been limited since 2012, when Hurricane Sandy breached Fire Island and re-opened the "Old Inlet". (*NY Sea Grant | Superstorm Sandy: One Year Later - Fire Island Breach*, n.d.).Understanding when and where access routes become compromised is essential for emergency planning and evacuation strategies.

Methods

The boundary of Fire Island, New York, was delineated using Google Earth Pro. The island perimeter was manually digitized using the polygon tool to accurately capture the spatial extent of the study area. The resulting polygon was exported as a shapefile (.shp) and used as the primary boundary layer for subsequent spatial analyses. A bare-earth Digital Elevation Model (DEM) was generated using 10-meter resolution National Elevation Dataset (NED) data. Elevation data were imported into Global Mapper, where they were clipped to the Fire Island shapefile boundary to isolate the study area. The DEM represents ground surface elevations with vegetation and built structures removed, allowing for accurate topographic analysis relevant to flood simulation. The 10 m NED DEM was used for island-wide analysis, while higher-resolution LiDAR-derived DEM data were used for detailed assessment of the Robert Moses Causeway and its bridge access infrastructure connecting Fire Island to Long Island.

The use of bare-earth LiDAR and DEM-based modeling follows established GIS approaches for flood hazard assessment, where high-resolution elevation data are used to simulate water-level thresholds and identify areas of potential inundation (Marsellos et al., 2010; Foster et al., 2011; Foster & Marsellos, 2012). Previous studies have demonstrated that removing vegetation and surface structures improves terrain representation and allows more accurate identification of flood-prone zones, particularly in low-relief environments (Marsellos, 2012; Rienzo et al., 2018). These methods have been widely applied in both ArcGIS and related geospatial platforms to reconstruct historical flooding and evaluate future risk scenarios.

A bathtub inundation model was implemented in Global Mapper, where areas with elevations below specified water surface levels and hydraulically connected to adjacent water bodies were classified as flooded. This elevation-threshold approach is consistent with previous GIS-based flood simulations in which water levels are incrementally raised across DEM surfaces to determine inundation extent and flood propagation patterns (Marsellos et al., 2010; Foster & Marsellos, 2012). The method has been applied in both ArcGIS and Global Mapper environments and is particularly suitable for barrier island settings such as Fire Island, where low relief and direct hydrologic connectivity to both the Atlantic Ocean and Great South Bay allow rapid lateral expansion of floodwaters.

Flooded cells were identified using an elevation-threshold rule, where a cell was classified as inundated if its ground elevation was less than or equal to the modeled water-surface elevation and if it was hydraulically connected to a water body, as expressed in the simplified formulation below.

$$I(x,y) = \begin{cases} 1, & \text{if } z(x,y) \leq H \text{ and } (x,y) \text{ is hydraulically connected} \\ 0, & \text{otherwise} \end{cases}$$

Where:

- $I(x,y)$ = inundation status at location (x,y)
- $z(x,y)$ = ground elevation from the DEM
- H = modeled water-surface elevation
- 1 = flooded cell
- 0 = non-flooded cell

Island-Wide Flood Simulation

Flood simulations were conducted across the entirety of Fire Island using the 10 m resolution NED bare-earth DEM within Global Mapper. Water surface elevation scenarios were modeled at 1 m, 2 m, 3 m, and 5 m above mean sea level. For each scenario, the water level was incrementally raised to simulate potential inundation extents. The resulting flooded areas were visually rendered, and flood extent maps were generated and analyzed to evaluate spatial patterns of inundation under each water level condition. This stepwise elevation modeling approach follows established DEM-based flood simulation techniques used in previous studies to quantify flood extent and identify critical inundation thresholds (Marsellos et al., 2010; Tsakiri et al., 2014).

Focused Analysis: West End Bridge Access (Robert Moses Beach)

To evaluate potential impacts to critical infrastructure, additional flood extent maps were generated for each water level scenario focusing specifically on the west end bridge access area near Robert Moses Beach. These images were used to assess vulnerability of transportation access under increasing flood conditions. Emphasis was placed on the west end access of the

Robert Moses Causeway over the east end access of the Smith's Point Bridge, due to the vulnerable state of the " Old Inlet" breach, which has caused limited westward access following Hurricane Sandy.

High-Resolution LiDAR-Based Flood Simulation

To enable higher-resolution analysis of the bridge access area, a LiDAR-derived bare-earth DEM was obtained and imported into Global Mapper. The LiDAR dataset, characterized by substantially finer spatial resolution than the 10 m NED data, was clipped to the bridge access study area.

Flood simulations were then performed on the LiDAR DEM at 1 m, 2 m, 3 m, and 5 m water surface elevations using the same methodology applied to the island-wide model. Flood extent maps of each simulation were captured to allow direct comparison between the lower-resolution NED-based simulations and the higher-resolution LiDAR-based results. The comparison between coarse and high-resolution DEMs is critical, as previous studies have shown that LiDAR-derived datasets significantly improve the detection of subtle elevation differences that control flood behavior in low-relief terrains (Foster & Marsellos, 2012; Weinstein & Marsellos, 2018).

Results

The spatial analysis reveals clear patterns in the progression of flooding across Fire Island. Low-lying bayside regions and interior sections between Robert Moses Beach and Watch Hill are among the first areas to become inundated under lower water level scenarios (1–2 m). In contrast, higher elevation dune systems along the Atlantic shoreline remain relatively protected during initial stages of flooding but become increasingly vulnerable at higher water levels (3–5 m), (see Figure 3), which illustrates the progressive inundation pattern across the island and highlights the early flooding of bayside and interior regions compared to the more resistant dune systems along the Atlantic shoreline.

Table 1. Total flooded area (km²) and percentage of inundation across Fire Island under incremental water level scenarios (1–5 m) based on bathtub model simulations.

m	Total area flooded (sq.km)	Percentage (%)
0	0	
1	11.59	45.0
2	17.31	67.1
3	21.04	81.6
4	23.02	89.3
5	24.4	94.6
Total area	25.78	

Road infrastructure is particularly sensitive to rising water levels. At approximately 1–2 m of water elevation, sections of low-lying access roads begin to experience inundation, especially in areas adjacent to the Great South Bay. As water levels increase to 2–3 m, connectivity along the island becomes increasingly fragmented, with multiple segments of the road network becoming impassable, (see Figure 4) where the central road corridor is shown to become increasingly fragmented as floodwaters expand from both the bayside and ocean-facing sides.

Of particular importance is the Robert Moses Causeway, which serves as the primary land-based access point to Fire Island. The analysis indicates that under moderate to high water level scenarios (approximately 2–3 m), portions of the approach to the causeway and surrounding infrastructure are at risk of flooding. At higher water levels (3–5 m), access to and from Long Island via the bridge is likely to be significantly disrupted or entirely compromised. This pattern is clearly illustrated in the high-resolution analysis of the Robert Moses area (see Figure 5), where floodwaters progressively encroach on access roads, parking areas, and critical approach routes to the causeway.

The results also demonstrate a clear sequence of infrastructure failure. Flooding initiates in low-lying bayside zones, followed by approaching into interior regions and secondary road segments. As water levels approach 2–3 m, the main transportation corridor becomes discontinuous, and access to critical infrastructure, including the Robert Moses Causeway, becomes unreliable. This progression highlights that functional accessibility is lost before complete physical inundation of the island occurs, emphasizing the importance of identifying operational thresholds rather than relying solely on total flood extent.

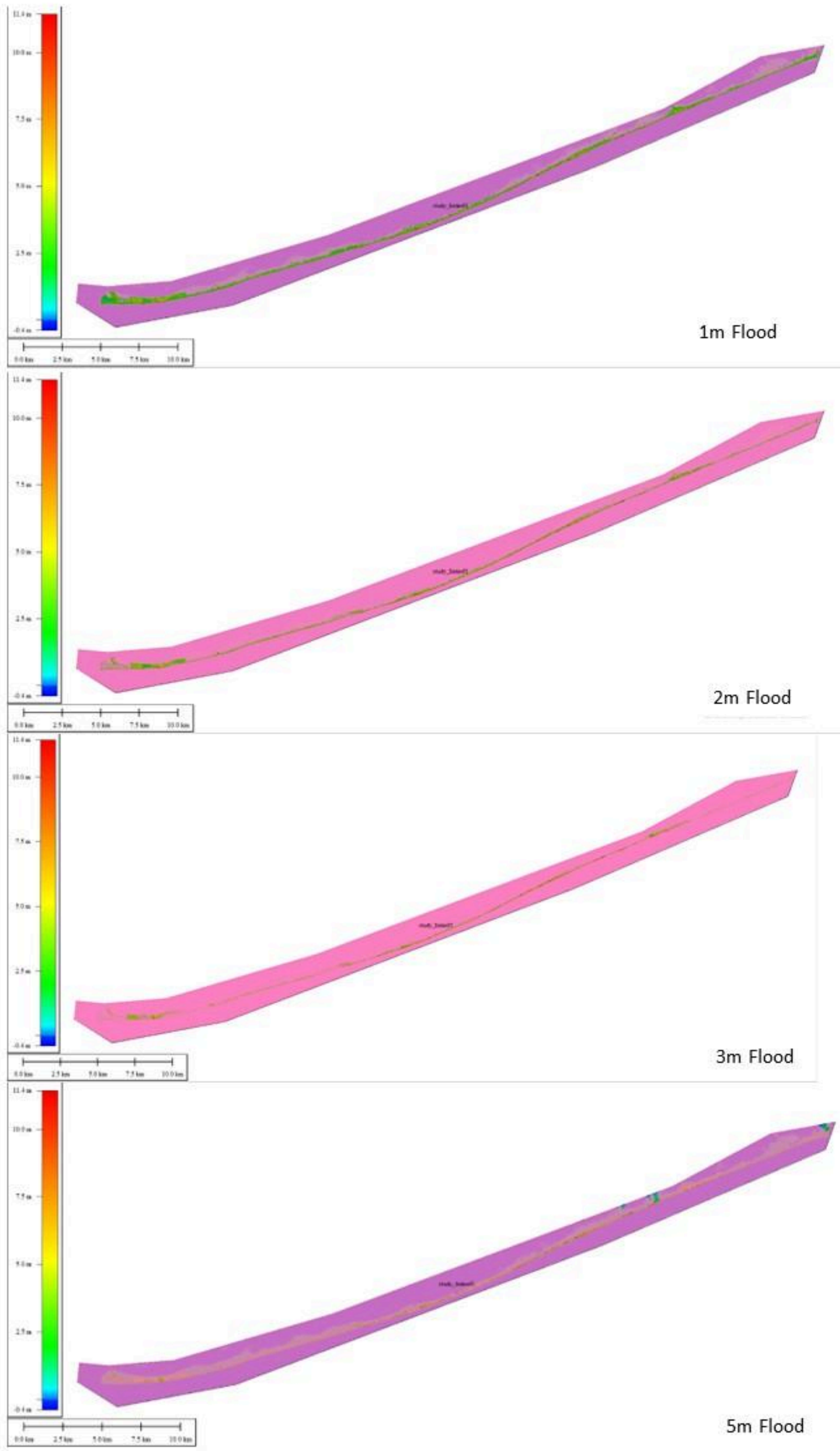

Figure 3: Flood Extent Maps, Robert Moses to Watch Hill

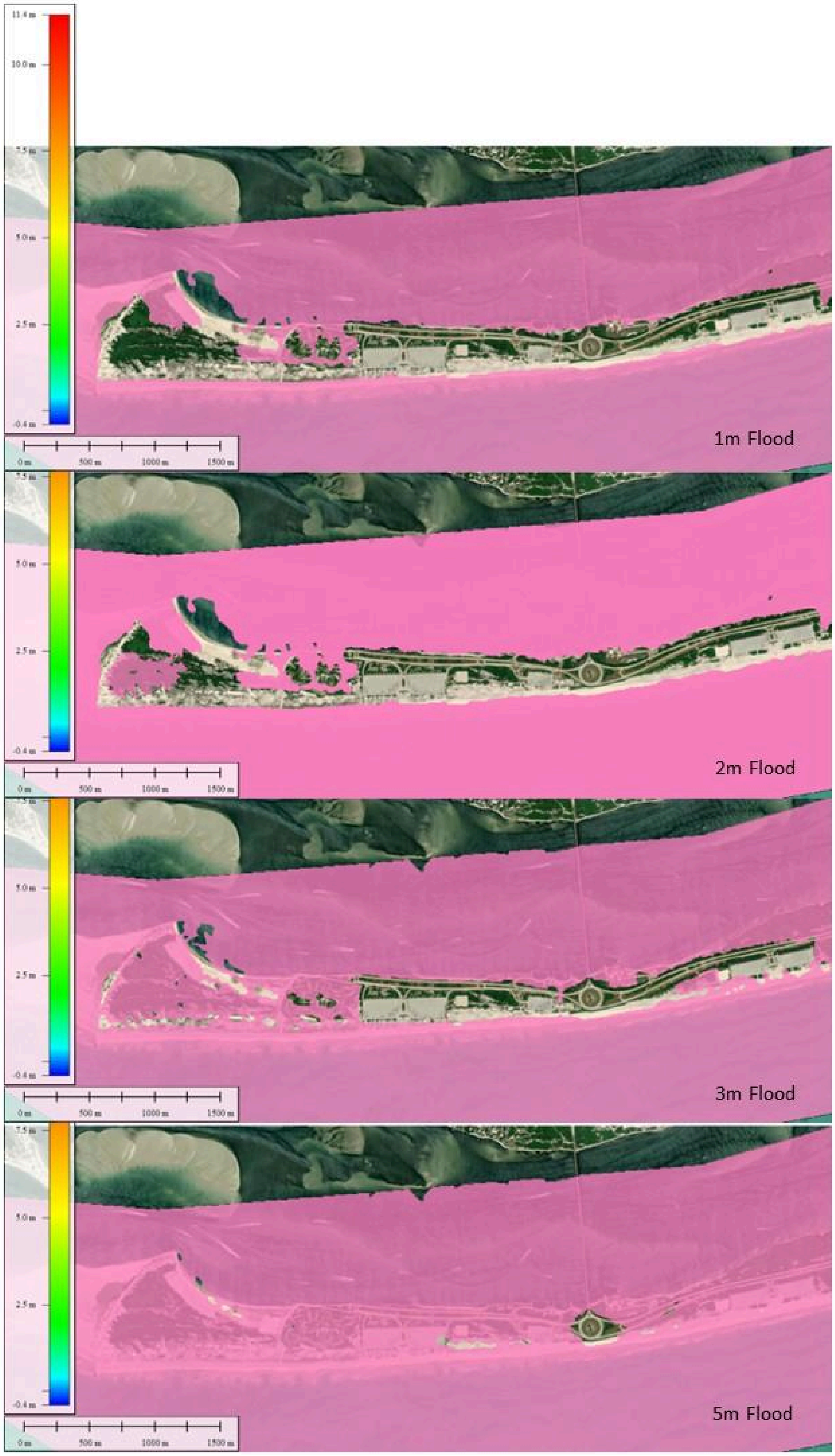

Figure 4: Flood Extent Maps of Robert Moses Bridge & Western Fire Island Road Network

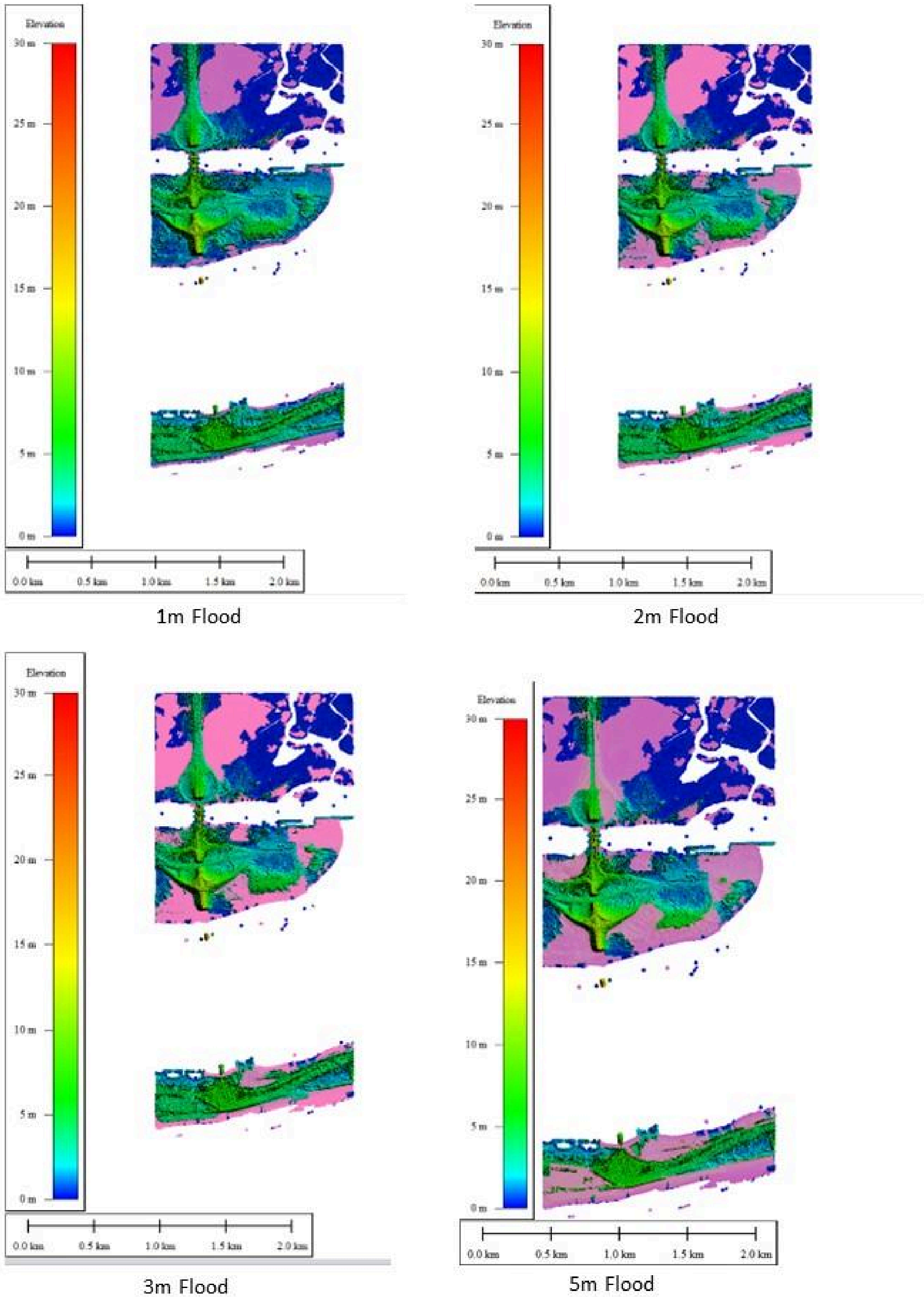

Figure 5: LiDAR Flood Extent Maps of Robert Moses Bridge

Discussion

The results of this study demonstrate that flood risk on Fire Island is not only a matter of spatial extent but also of critical infrastructure vulnerability. The application of the bathtub model highlights how even modest increases in water level can lead to early inundation of low-lying interior and bayside areas, followed by progressive disruption of transportation routes.

One of the most significant findings of this study is that infrastructure failure occurs prior to complete physical inundation of the landscape. The progressive fragmentation of the road network, particularly along the central corridor and near the Robert Moses Causeway, indicates that functional accessibility is lost at intermediate water levels (~2–3 m). This distinction between physical inundation and functional failure is critical. Emergency response and evacuation planning often rely on assumptions about total flood extent; however, this study demonstrates that transportation networks may become impassable well before the island is fully submerged. As a result, identifying operational thresholds for infrastructure failure is more relevant for hazard mitigation than simply mapping maximum inundation boundaries.These findings underscore the importance of integrating flood modeling with transportation and emergency planning. Identifying thresholds at which access routes fail provides valuable insight for evacuation planning, emergency response coordination, and infrastructure resilience in coastal environments.

Future research should extend the LiDAR-based flood simulation framework to the eastern access point of Fire Island at the Smith Point Bridge, which serves as the only alternative vehicular connection to the island. While this study focused on the vulnerability of the Robert Moses Causeway, a comprehensive assessment of island-wide accessibility requires evaluation of both primary access routes.

High-resolution LiDAR-derived DEMs should be utilized to model flood scenarios in the Smith Point region under the same incremental water-level conditions (1–5 m). This analysis would enable direct comparison between the western and eastern access points, identifying differences in elevation, flood propagation, and infrastructure resilience.

Additionally, the study should assess the timing and sequence of roadway inundation leading to potential isolation of Fire Island. By integrating both access points into a unified framework, future work can provide a more complete understanding of evacuation feasibility and critical thresholds at which the island becomes fully inaccessible.

Such analysis would further support emergency planning, infrastructure design, and coastal resilience strategies by identifying whether redundancy exists between access routes or if both are vulnerable under similar flood conditions.

Conclusion

This study shows that flood risk on Fire Island is strongly controlled by subtle elevation differences and the connectivity between bayside, interior, and ocean-facing zones. GIS- and LiDAR-based modeling indicates that low-lying areas are inundated under relatively modest water levels (1–2 m), while dune systems provide only limited protection under higher scenarios.

A critical finding is that infrastructure failure occurs before full inundation. The progressive fragmentation of the road network and disruption of access—especially near the Robert Moses Causeway—suggest that functional accessibility is lost at intermediate water levels (~2–3 m), highlighting the importance of identifying operational thresholds for emergency planning.

The bathtub-style inundation model, combined with high-resolution DEMs, proved effective for evaluating flood propagation in this low-relief barrier island setting. These results provide practical insight for coastal management and infrastructure planning and demonstrate a transferable framework for assessing flood vulnerability in similar coastal environments.

Credit Authorship Contribution Statement

Runkel, M.: Writing- Abstract, Intro, Methods, Results, Discussion, Conclusion, and References.editing, literature, global mapper flood simulations, Figure 1, Figure 2, Figure 3, Figure 4, Figure 5. Marsellos, A.E.: supervision, guidance, editing.

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